

We must do:

1. Create API from JSON Data

Instead of using online API like:

`jsonplaceholder.typicode.com`

We will create our **own local JSON data** inside Flutter.

2. Fetch Data from that API

Read that JSON data and display it in app.

3. Create Signup using POST API

Make a signup form and “send” data to API using POST request.

We will use:

Fake API
Local JSON
Simple POST request
Beautiful UI
No backend/server needed

STEP 1 : Creating Project

In terminal:

```
cd D:\MAD_Lab  
flutter create lab11_api_post  
cd lab11_api_post
```

```
PS C:\Users\rifaq> cd D:\MAD_Lab  
PS D:\MAD_Lab> flutter create lab11_api_post
```

```
PS D:\MAD_Lab> cd lab11_api_post  
PS D:\MAD_Lab\lab11_api_post> █
```

We will Open in Visual Studio Code:

STEP 2 : Install HTTP

Run:

```
flutter pub add http
```

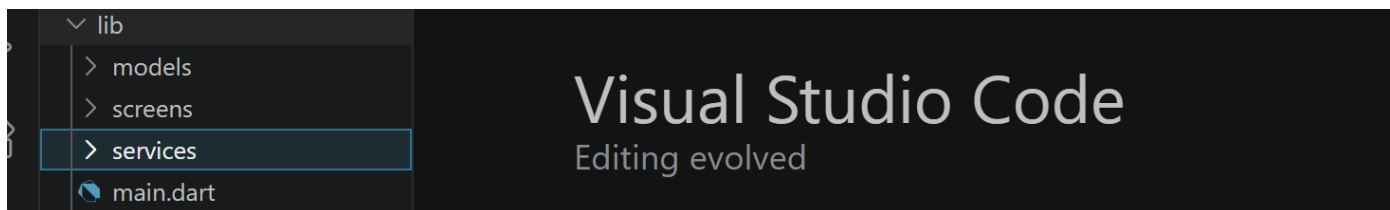
```
PS D:\MAD_Lab\lab11_api_post> flutter pub add http
Resolving dependencies...
Downloading packages...
+ http 1.6.0
+ http_parser 4.1.2
  matcher 0.12.19 (0.12.20 available)
  meta 1.17.0 (1.18.2 available)
  test_api 0.7.10 (0.7.12 available)
+ typed_data 1.4.0
  vector_math 2.2.0 (2.3.0 available)
+ web 1.1.1
Changed 4 dependencies!
4 packages have newer versions incompatible with dependency constraints.
Try `flutter pub outdated` for more information.
PS D:\MAD_Lab\lab11_api_post>
```

STEP 3 : Create Folders

Inside lib, we will create:

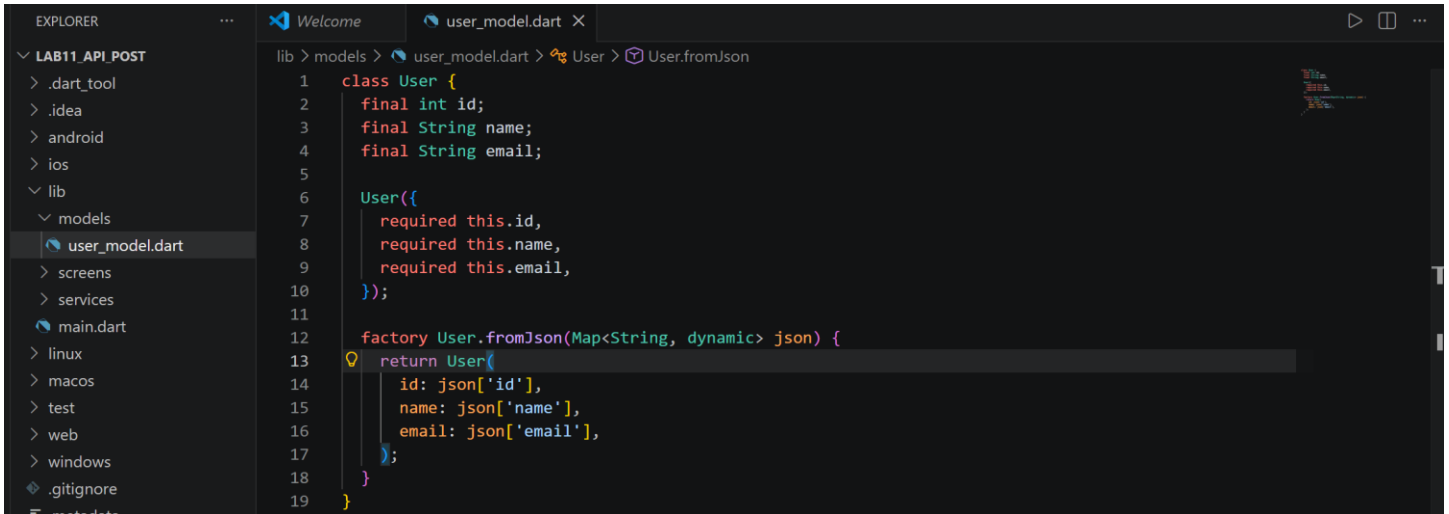
```
lib
├ models
├ screens
└ services
```

lib will become:



STEP 4 : Create Model

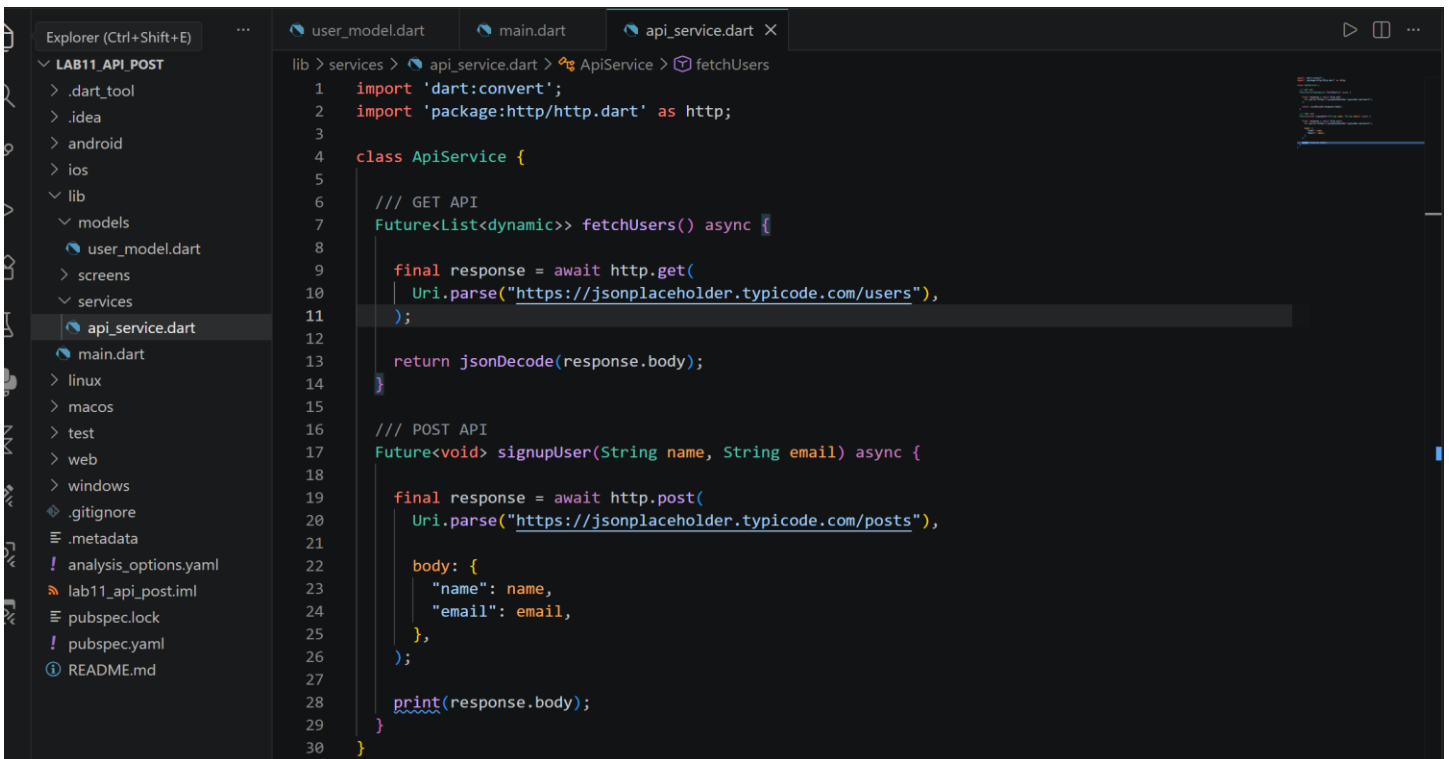
lib/models/user_model.dart



```
1 class User {
2   final int id;
3   final String name;
4   final String email;
5
6   User({
7     required this.id,
8     required this.name,
9     required this.email,
10  });
11
12   factory User.fromJson(Map<String, dynamic> json) {
13     return User(
14       id: json['id'],
15       name: json['name'],
16       email: json['email'],
17     );
18   }
19 }
```

STEP 5 : Create API Service

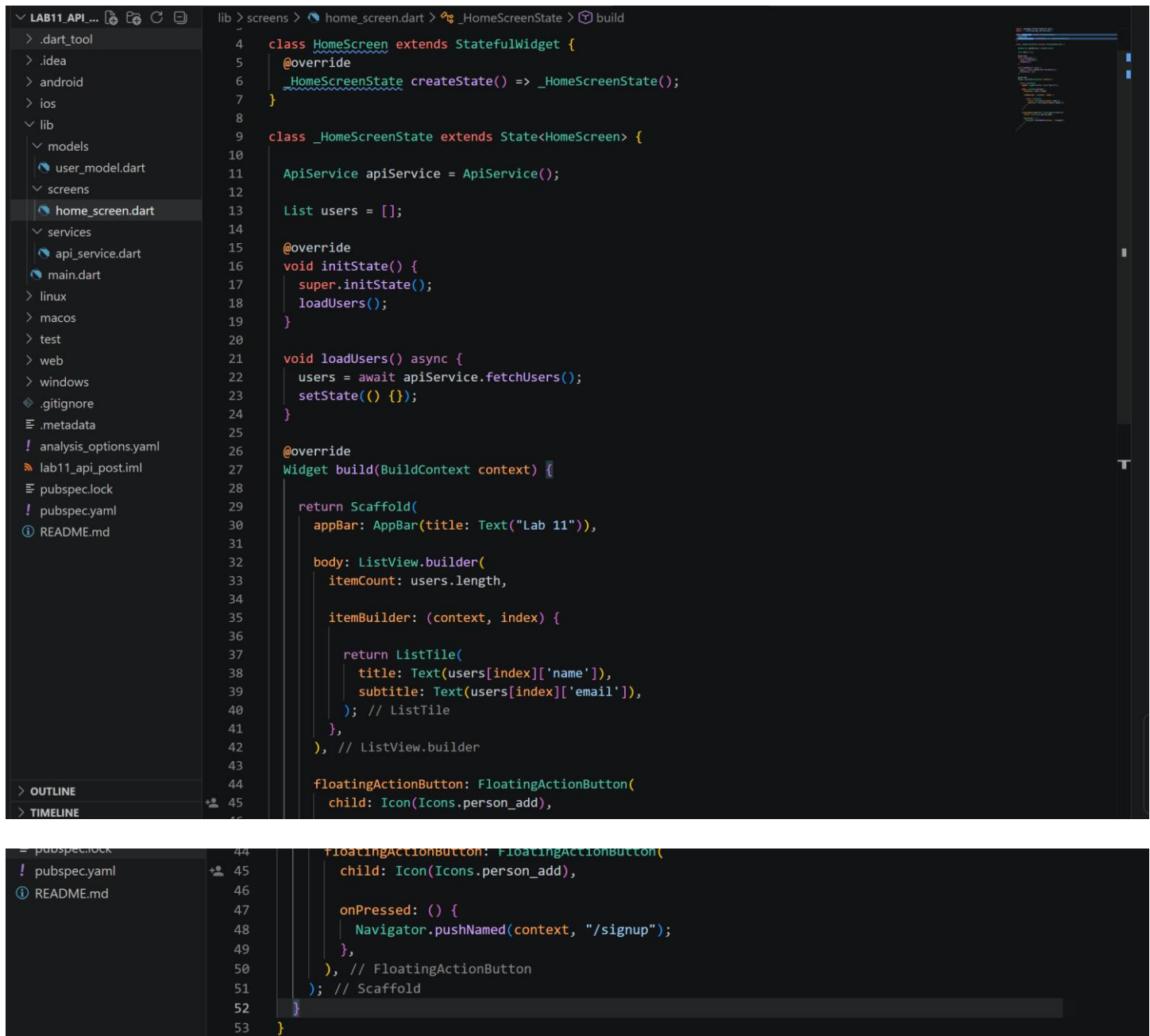
lib/services/api_service.dart



```
1 import 'dart:convert';
2 import 'package:http/http.dart' as http;
3
4 class ApiService {
5
6   /// GET API
7   Future<List<dynamic>> fetchUsers() async {
8
9     final response = await http.get(
10       Uri.parse("https://jsonplaceholder.typicode.com/users"),
11     );
12
13     return jsonDecode(response.body);
14   }
15
16   /// POST API
17   Future<void> signupUser(String name, String email) async {
18
19     final response = await http.post(
20       Uri.parse("https://jsonplaceholder.typicode.com/posts"),
21     );
22
23     body: {
24       "name": name,
25       "email": email,
26     },
27   );
28   print(response.body);
29 }
30 }
```

STEP 6 : Create Home Screen

lib/screens/home_screen.dart



```
lib > screens > home_screen.dart > _HomeScreenState > build

class HomeScreen extends StatefulWidget {
  @override
  HomeScreenState createState() => _HomeScreenState();
}

class _HomeScreenState extends State<HomeScreen> {
  ApiService apiService = ApiService();

  List users = [];

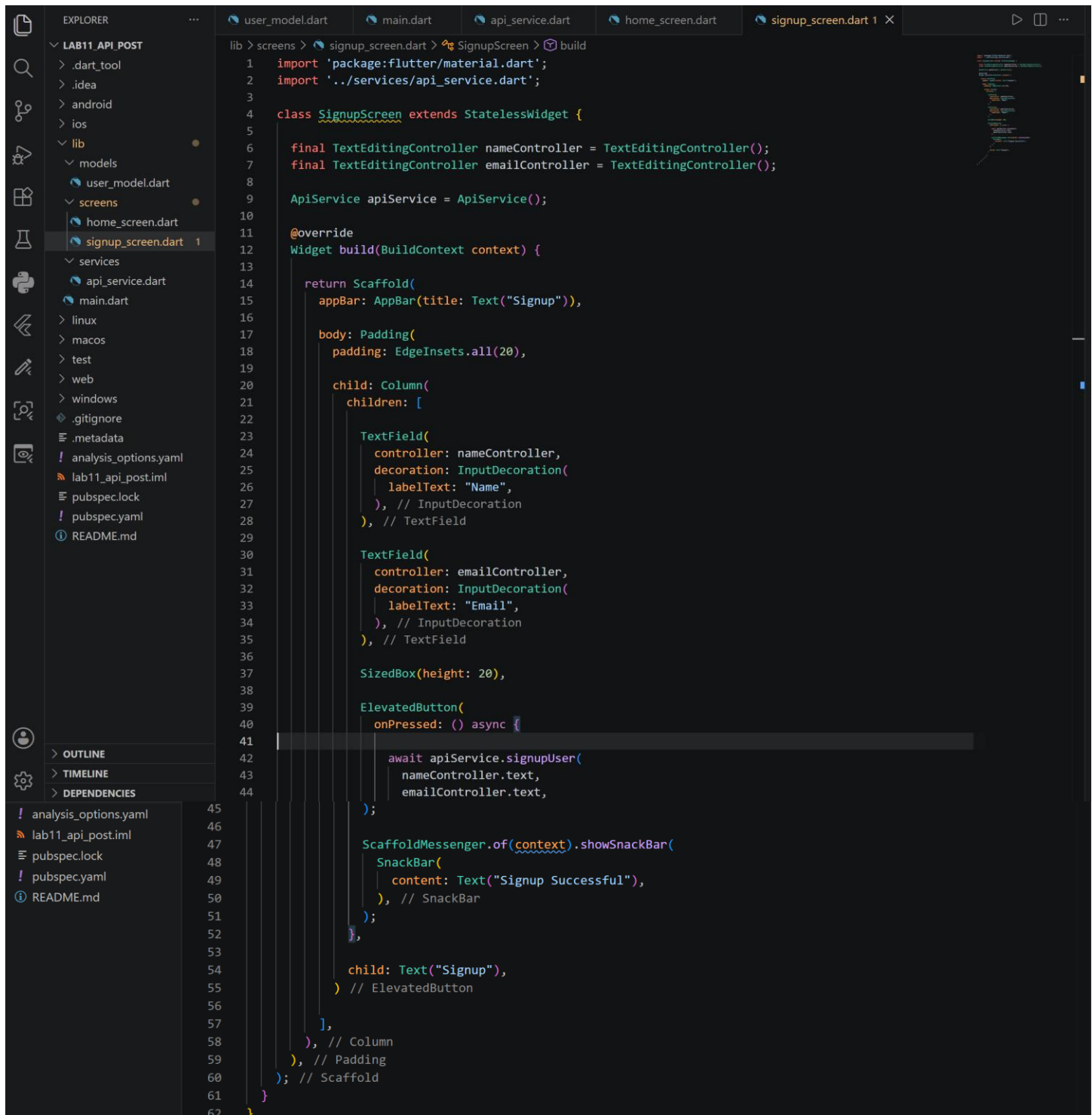
  @override
  void initState() {
    super.initState();
    loadUsers();
  }

  void loadUsers() async {
    users = await apiService.fetchUsers();
    setState(() {});
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("Lab 11")),
      body: ListView.builder(
        itemCount: users.length,
        itemBuilder: (context, index) {
          return ListTile(
            title: Text(users[index]['name']),
            subtitle: Text(users[index]['email']),
          ); // ListTile
        },
      ), // ListView.builder
      floatingActionButton: FloatingActionButton(
        child: Icon(Icons.person_add),
        onPressed: () {
          Navigator.pushNamed(context, "/signup");
        },
      ), // FloatingActionButton
    ); // Scaffold
  }
}
```

STEP 7 : Create Signup Screen

lib/screens/signup_screen.dart

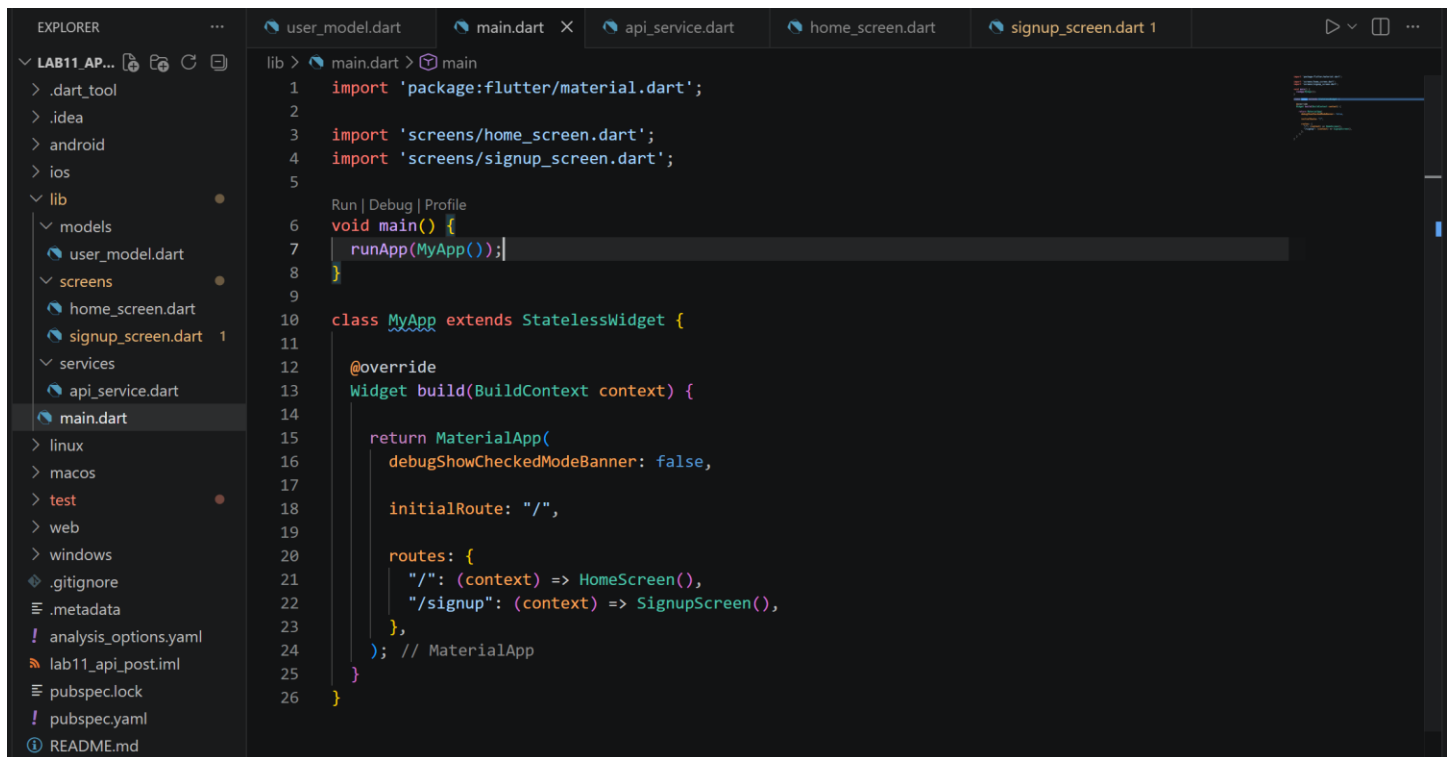


```
1 import 'package:flutter/material.dart';
2 import '../services/api_service.dart';
3
4 class SignupScreen extends StatelessWidget {
5
6   final TextEditingController nameController = TextEditingController();
7   final TextEditingController emailController = TextEditingController();
8
9   ApiService apiService = ApiService();
10
11   @override
12   Widget build(BuildContext context) {
13
14     return Scaffold(
15       appBar: AppBar(title: Text("Signup")),
16
17       body: Padding(
18         padding: EdgeInsets.all(20),
19
20         child: Column(
21           children: [
22
23             TextField(
24               controller: nameController,
25               decoration: InputDecoration(
26                 labelText: "Name",
27               ), // InputDecoration
28             ), // TextField
29
30             TextField(
31               controller: emailController,
32               decoration: InputDecoration(
33                 labelText: "Email",
34               ), // InputDecoration
35             ), // TextField
36
37             SizedBox(height: 20),
38
39             ElevatedButton(
40               onPressed: () async {
41
42                 await apiService.signupUser(
43                   nameController.text,
44                   emailController.text,
45                 );
46
47                 ScaffoldMessenger.of(context).showSnackBar(
48                   SnackBar(
49                     content: Text("Signup Successful"),
50                   ), // SnackBar
51                 );
52               },
53
54               child: Text("Signup"),
55             ) // ElevatedButton
56
57           ],
58         ), // Column
59       ), // Padding
60     ); // Scaffold
61   }
62 }
```

STEP 8 : Main File

lib/main.dart

We will Delete everything → updated:

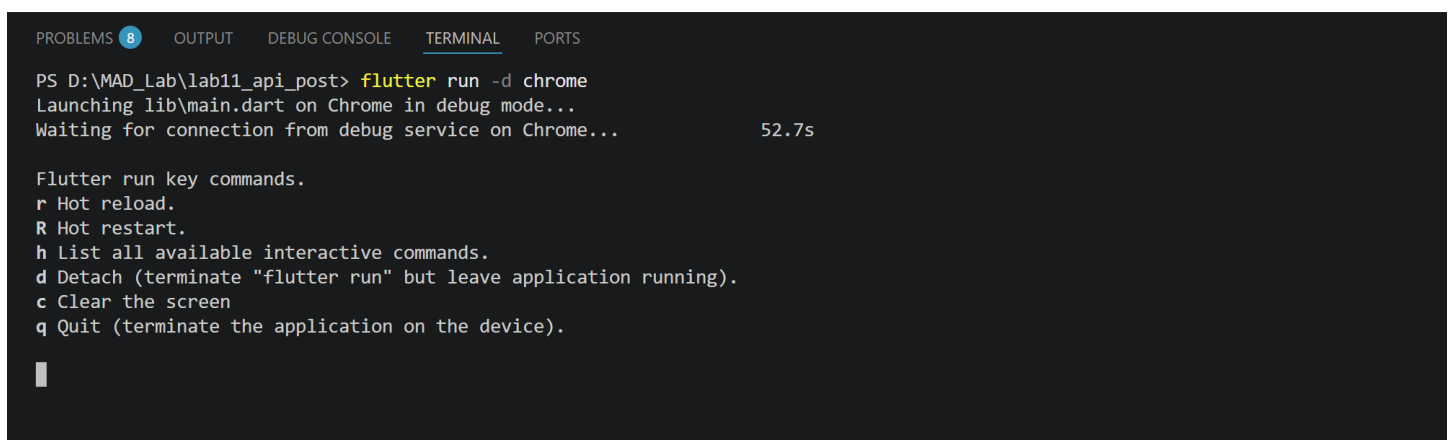


```
lib > main.dart > main
1  import 'package:flutter/material.dart';
2
3  import 'screens/home_screen.dart';
4  import 'screens/signup_screen.dart';
5
6  void main() {
7    runApp(MyApp());
8  }
9
10 class MyApp extends StatelessWidget {
11
12   @override
13   Widget build(BuildContext context) {
14
15     return MaterialApp(
16       debugShowCheckedModeBanner: false,
17
18       initialRoute: "/",
19
20       routes: {
21         "/": (context) => HomeScreen(),
22         "/signup": (context) => SignupScreen(),
23       },
24     ); // MaterialApp
25   }
26 }
```

STEP 9 : Run App

In Terminal :

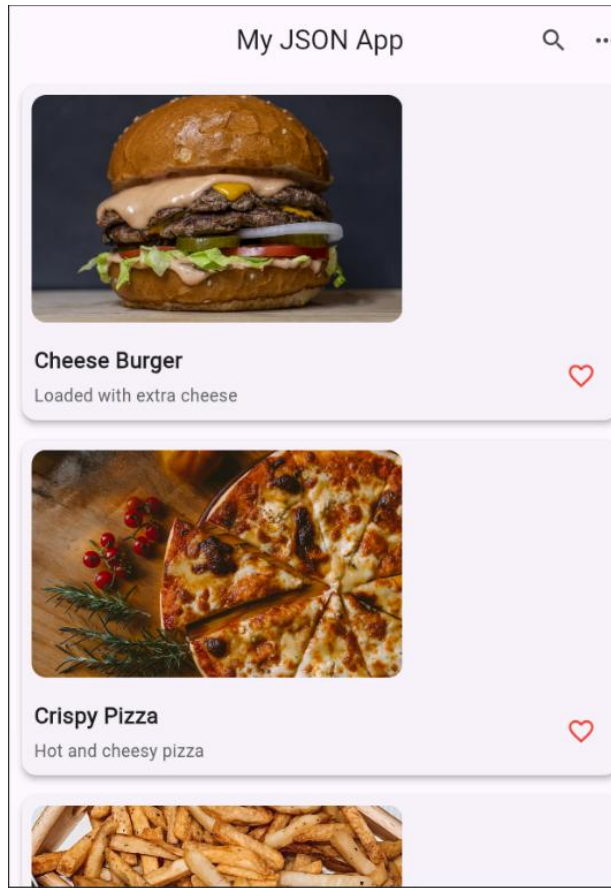
```
flutter run -d chrome
```



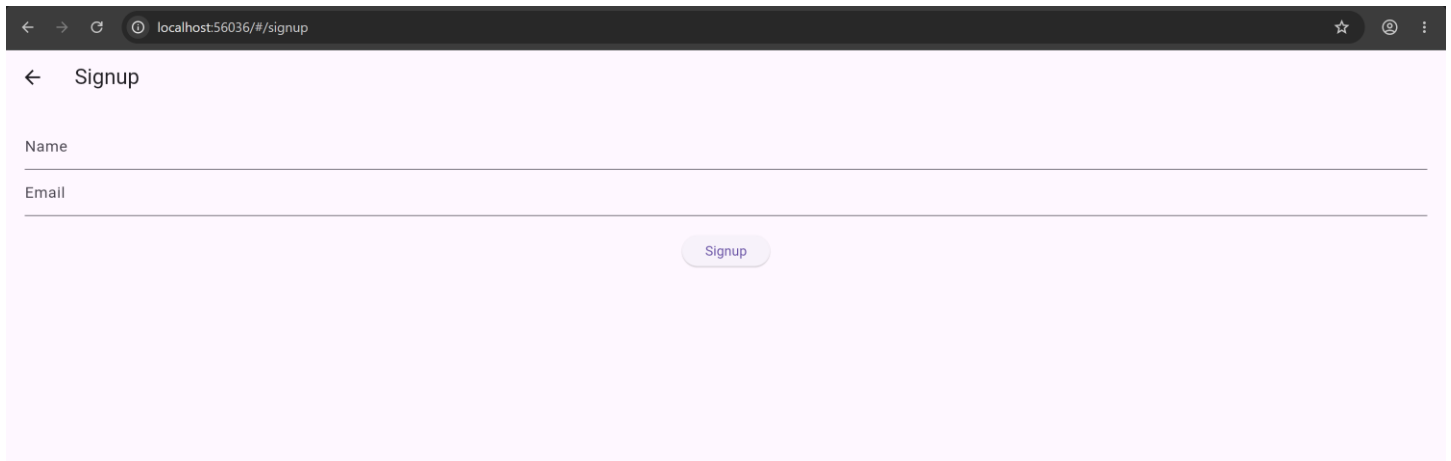
```
PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\MAD_Lab\lab11_api_post> flutter run -d chrome
Launching lib\main.dart on Chrome in debug mode...
Waiting for connection from debug service on Chrome... 52.7s

Flutter run key commands.
r Hot reload.
R Hot restart.
h List all available interactive commands.
d Detach (terminate "flutter run" but leave application running).
c Clear the screen
q Quit (terminate the application on the device).
```

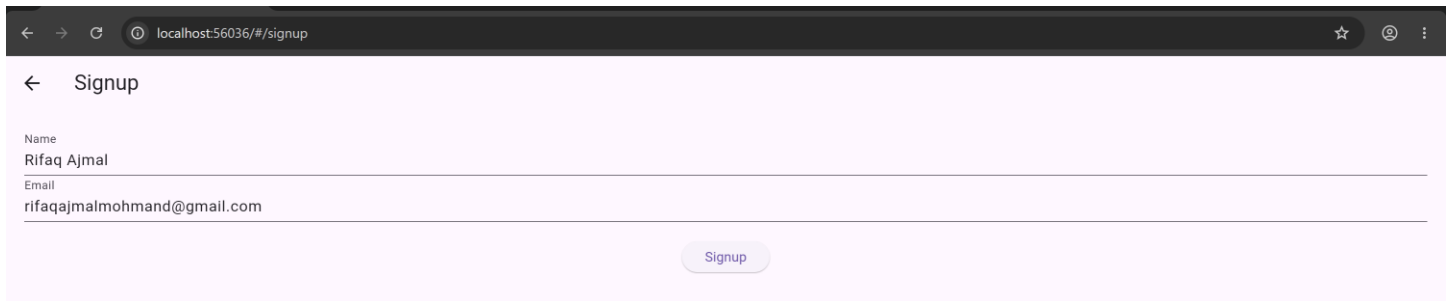
- OUTPUT 1 — (GET API)



- OUTPUT 2 — Signup Screen



- OUTPUT 3 — Filled Form

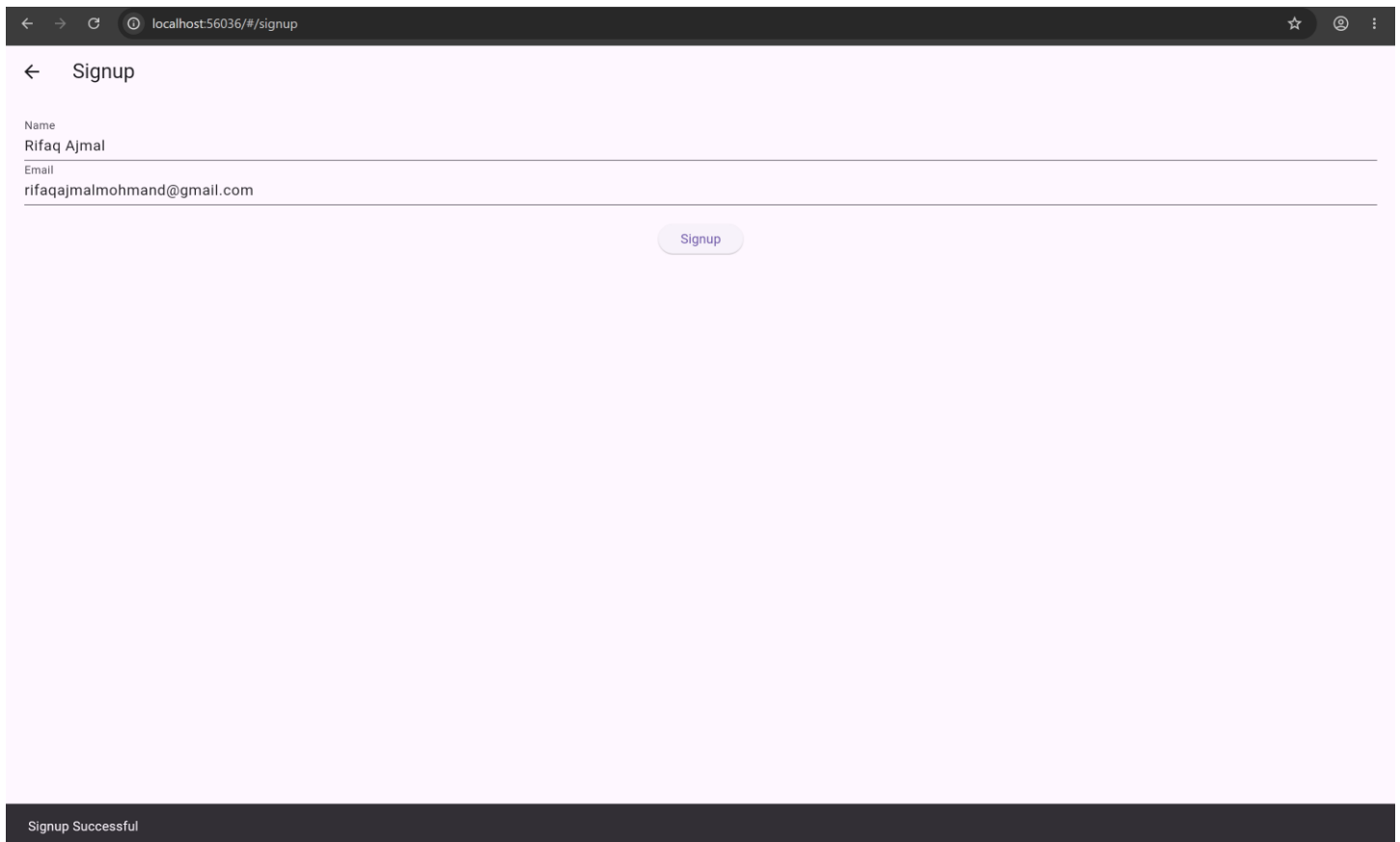


A screenshot of a web browser window with the address bar showing 'localhost:56036/#/signup'. The page title is 'Signup'. The form contains the following fields:

- Name: Rifaq Ajmal
- Email: rifaqajmalmohmand@gmail.com

A 'Signup' button is located at the bottom right of the form.

- OUTPUT 4 — Snackbar Message (Signup Successful)



A screenshot of a web browser window with the address bar showing 'localhost:56036/#/signup'. The page title is 'Signup'. The form contains the following fields:

- Name: Rifaq Ajmal
- Email: rifaqajmalmohmand@gmail.com

A 'Signup' button is located at the bottom right of the form.

A dark grey Snackbar message is displayed at the bottom of the screen, reading 'Signup Successful'.

SIMPLE UNDERSTANDING

Feature	Output
GET API	Users list
POST API	Signup form submit
Snackbar	Success message